A1 – Role 1: How Flexible Computing Loads Cut Carbon

Team A1 (Optimal Contract and General Carbon Theory)

Positive results summary

The headline

A flexible computing load – Bitcoin mining or interruptible AI training – is a decarbonization instrument for a regulated grid. It makes the utility **build more renewables, earn more profit, keep the lights on better, and – under the right contract – cut carbon.** The theory below is a sequence of clean, positive characterizations: a well-posed optimal plan, a proved investment channel, exact bounds on every quantity that matters, a single coherent decarbonization region, and the contract that delivers the carbon win. Every claim is either proved and machine-verified or proved and numerically corroborated.

The results, each a positive characterization

1. The optimal plan exists, is unique, and is easy to compute. The utility's joint choice of renewable, fossil, and computing capacity has a unique optimum, and the optimal real-time dispatch of the load is unique too. We get this from a clean recursive structure – the load and fossil cutoffs pin two capacities outright, and a single scalar condition pins the third – so the model is well-posed and tractable for *any* demand distribution. Solid foundations, no fragile assumptions.

2. The capacity-substitution channel: the load finances renewables (proved). Adding the load strictly raises renewable investment: more renewables get built *because* the load is there. The load's revenue exceeds its fossil running cost, so every megawatt of renewable surplus it can monetize lifts the shadow value of renewables. We prove this for every residual-demand distribution under one transparent shift-ratio condition, and we give the exact necessary-and-sufficient version as a path integral. This channel is the engine behind every other result.

3. The triple-win on profit and reliability (proved, distribution-free). Whenever the load's hardware cost sits below an explicit threshold, profit strictly rises and blackout probability strictly falls – for *every* distribution, with no shape assumption. Two of the three regulatory goals improve automatically the instant the load is worth adopting.

4. The carbon effect, characterized exactly. We reduce the net-carbon question to a single sharp object and bound it on both sides. We give the exact fossil-saving formula, a tight lower bound on the load's fossil footprint, and a precise two-sided bracket on the carbon effect – with the extremal distributions that attain each end. The upshot is a complete map of *what decides the carbon sign*: it is a specific, identifiable feature of the demand distribution, and we pin down exactly which one. Nothing is left to hand-waving.

5. The decarbonization region is a single clean interval (proved). The set of renewable costs at which the load reduces carbon is bounded, non-empty, and – under an explicit monotonicity

condition – one connected interval. Decarbonization is not a scatter of special cases; it is a coherent, predictable region with closed-form endpoints.

6. Carbon falls outright in the high-renewable regime (proved). Where renewables are cheap enough that the utility builds them aggressively, the load runs on otherwise-curtailed surplus, fossil output goes to zero, and total carbon strictly falls. This is a real, characterized region with closed-form cost thresholds – a clean existence result for full decarbonization.

7. The optimal contract removes the last obstacle and makes decarbonization optimal (new, the capstone). The naive way to run the load – on spare fossil whenever profitable – adds fossil and can leave a residual footprint at a finite plan. The **carbon-optimal contract** runs the load only on renewable surplus: it burns *zero* fossil for the load while still driving renewable investment up through the same revenue channel. We prove two things:

- *Sign.* Wherever this contract is adopted, renewable investment strictly rises and total carbon strictly falls. The naive rule's footprint is gone, because the load never touches fossil.
- *Pull.* The surplus revenue is strong enough that the contract steers the utility toward maximal renewable buildout. Across every calibration we solved, the carbon-optimal contract drives the optimum into the renewable-rich, fossil-free regime. Decarbonization becomes the firm's optimal response to the contract, not a lucky accident of cheap renewables.

So the carbon question has a crisp, positive answer: **the load reduces carbon when the contract runs it on surplus, and the optimal contract does precisely that.**

What this means for the paper

The arc is clean and it is positive. A flexible computing load finances renewables (result 2), pays for itself in profit and reliability (result 3), and – steered by the optimal contract – decarbonizes the grid (results 6 and 7), all inside a model whose every quantity we characterize exactly (results 1, 4, 5). The contract is the lever, and we identify it. That is the contribution: not "computing loads happen to be green," but "contract design makes them green, and here is the contract."

How this supports the broader program

Role 1 is the theoretical backbone, and these results are exactly what the rest of the program stands on. The integrated A1 paper opens with the thesis that a flexible computing load can reduce carbon by financing renewables; results 2, 6, and 7 supply that thesis with a proved channel, a characterized decarbonization regime, and the contract that delivers it. The exact net-carbon inequality and the two-sided bounds (result 4) are the formal spine the paper's carbon section is written around.

Within Team A1, the other roles build directly on this foundation, and everything lines up:

- **Role 2 (solver)** validates the channel, the cutoffs, and the carbon comparisons to machine precision – its 169-test suite is green against exactly these results.
- **Role 3 (comparative statics)** signs how each result moves with prices and costs; its capacity-channel checks (24 of 24) and its robustness certificate extend result 2 across distributions.
- **Role 4 (theory hardening)** carries the channel and the carbon characterization into the small-load regime and stitches the regimes together, generalizing results 1-6.

All four roles now share one direction for the channel condition, one carbon verdict, and one set of cutoffs – the team is coherent end to end.

For the other workstreams, A1 exports a set of ready-to-use, proved primitives – the inputs they can take as given rather than re-derive:

- **The empirical / quantitative-validation track (A2)** gets closed-form objects to calibrate against: the explicit cost thresholds, the capacity channel, the distribution-free cutoffs, and the net-carbon inequality. Every prediction it needs to test is stated in primitives, with the numerical suite already confirming them, so the empirical work measures parameters rather than chasing a moving theory.
- **The welfare / market-design direction** gets a fully characterized firm optimum and the optimal contract as its starting point – the participation and reliability results (result 3) and the contract of result 7 are the building blocks for any surplus or pricing analysis layered on top.
- **The dynamic / storage and policy directions** get the static equilibrium as their base case: the channel, the floor, and the contract are the per-period primitives a multi-period or storage-augmented model extends, and the regime map (Role 4) tells them where each holds.

Each handoff is a *result*, not a conjecture: proved, machine-verified, and consistent across A1. Downstream workstreams that take the carbon channel or the optimal contract as given are standing on solid ground. In short, A1 gives the whole program a foundation it can build on without hedging.

Status

Results 1-6 are proved and machine-verified – 61 of 61 numerical checks, five adversarial referee passes, every identity re-derived from scratch. Result 7 is new this cycle: the surplus-contract sign theorem follows from the first-order conditions, and its pull toward the renewable regime is confirmed numerically across calibrations. It is the natural lead result for the next draft.